Fields and Waves I

Lecture 15

Magnetostatics & Ampere's Law

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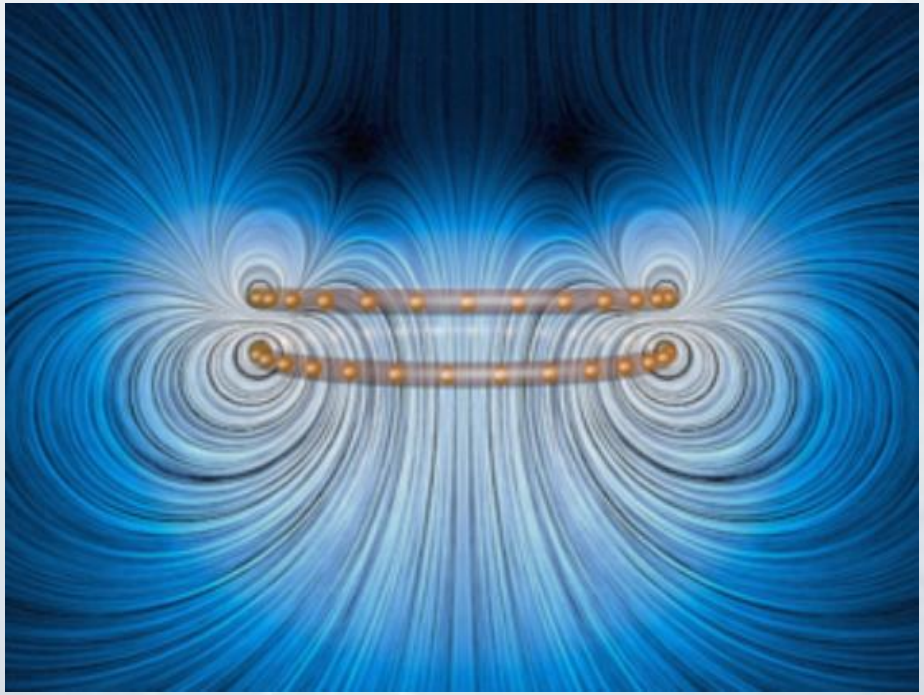
These slides were prepared through the work of the following people:

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Materials from other sources are referenced where they are used.
Those listed as Ulaby are figures from Ulaby's textbook.

Overview

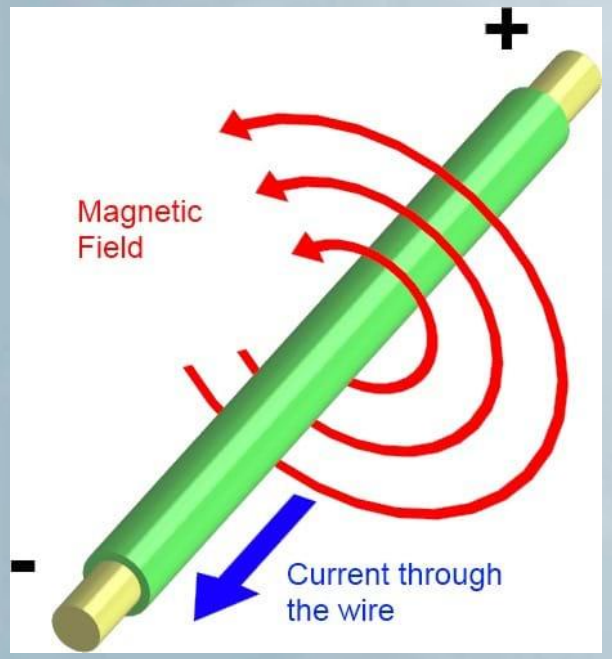
- Magnetostatics
- Ampere's Law



https://web.mit.edu/8.02t/www/802TEAL3D/visualizations/magnetostatics/MagneticForceRepel/MagneticForceRepel.htm

Magnetostatics

Magnetostatics



TeachEngineering.org

- In magnetostatics, we have current. All currents are constant.
- Thus, all magnetic fields are constant

Magnetostatics

Electrostatic Version of Maxwell's Equations

Integral Form	Differential Form
$\oint \vec{D} \cdot d\vec{S} = \int \rho dV$	$\nabla \cdot \vec{D} = \rho$
$\oint \vec{E} \cdot d\vec{l} = 0$	$\nabla \times \vec{E} = 0$

Magnetostatics

Magnetostatic Version of Maxwell's Equations

Integral Form	Differential Form
$\oint \vec{H} \cdot d\vec{l} = \int \vec{J} \cdot d\vec{s} = I_{enc}$	$\nabla \times \vec{H} = \vec{J}$
$\oint \vec{B} \cdot d\vec{S} = 0$	$\nabla \cdot \vec{B} = 0$

Magnetostatics

- Electrostatics is based on two of Maxwell's equations, but simplified.
- Magnetostatics is based on the other two Maxwell's equations, but simplified.
- In these simplified Maxwell's equations, electric and magnetic fields are separate. In the full Maxwell's equations below, they are coupled.

Integral Form	**Differential Form**
$\oint \vec{B} \cdot d\vec{S} = 0$	$\vec{\nabla} \cdot \vec{B} = 0$
$\oint \vec{D} \cdot d\vec{S} = \int \rho dV$	$\vec{\nabla} \cdot \vec{D} = \rho$
$\oint \vec{H} \cdot d\vec{l} = \int \vec{J} \cdot d\vec{S} + \frac{d}{dt} \int \vec{D} \cdot d\vec{S}$	$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$
$\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{S}$	$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

Magnetostatics

Magnetostatic Version of Maxwell's Equations

Integral Form	Differential Form
$\oint \vec{H} \cdot d\vec{l} = \int \vec{J} \cdot d\vec{s} = I_{enc}$	$\nabla \times \vec{H} = \vec{J}$
$\oint \vec{B} \cdot d\vec{S} = 0$	$\nabla \cdot \vec{B} = 0$

These equations tell you that magnetic field arises from current (and circulates around the current).

Magnetostatics

Magnetostatic Version of Maxwell's Equations

Integral Form	Differential Form
$\oint \vec{H} \cdot d\vec{l} = \int \vec{J} \cdot d\vec{s} = I_{enc}$	$\nabla \times \vec{H} = \vec{J}$
$\oint \vec{B} \cdot d\vec{S} = 0$	$\nabla \cdot \vec{B} = 0$

These equations tell you that magnetic field doesn't have any sources or sinks (i.e. no "magnetic charge".) It just circulates.

Magnetostatics

- There are 2 different ways of writing the magnetic field.

  $\vec{H}$ (often called just called "magnetic field" or "H-field") has units of amperes per meter (**A/m**). Thus it has a direct relationship with **current**.

  $\vec{B}$ (often called "magnetic flux density"), has units of teslas (**T**) or webers per square meter (**Wb/m²**).

- These two quantities are related by the **magnetic permeability** μ, which is a property of the magnetic material that the field is in

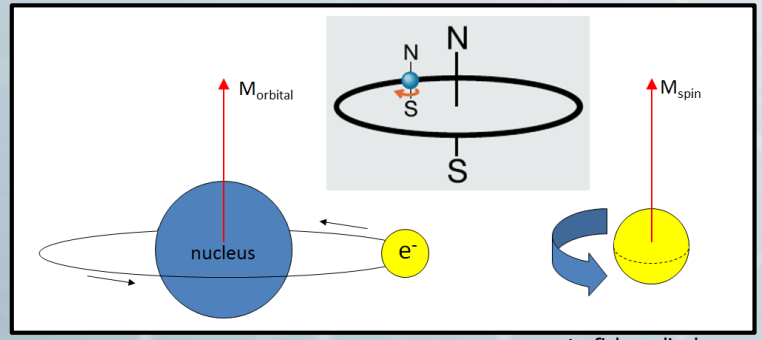
$$\vec{B} = \mu\vec{H}$$

$$\mu_0 = 4\pi \times 10^{-7}\ \mathrm{H/m}$$ ← Magnetic permeability of free space

- We will mostly deal with $\vec{B}$ because it is responsible for the most interesting physical consequences of magnetism

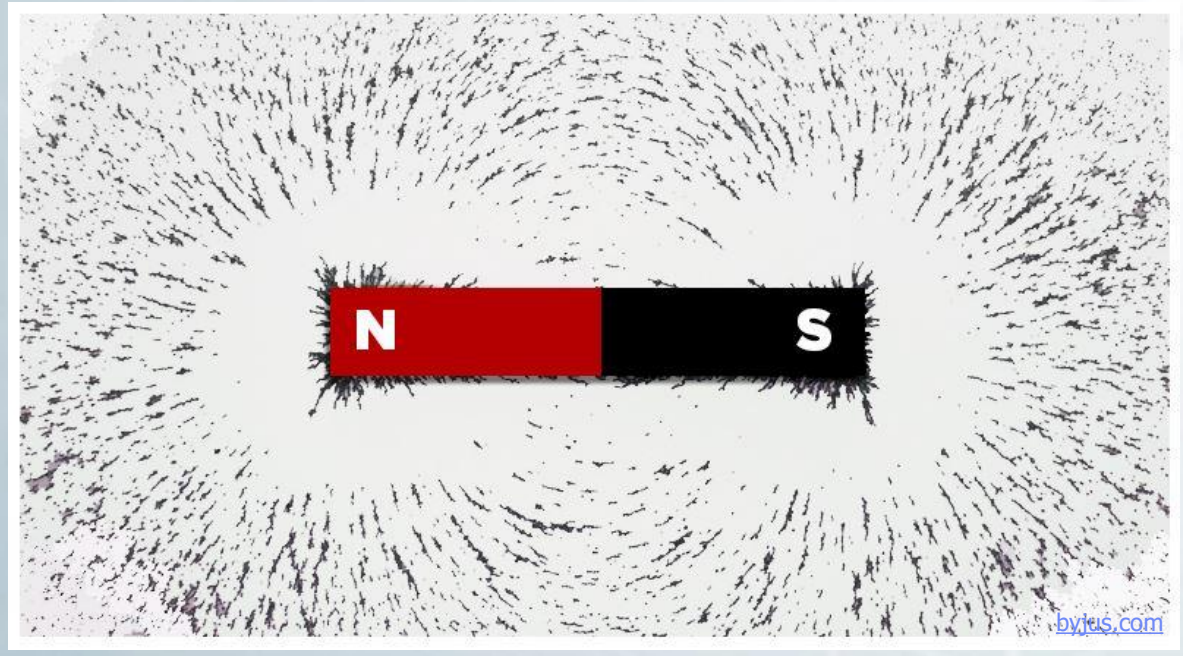
Magnetostatics

- What determines a material's permeability or whether or not it is magnetic?
  - Atoms are tiny magnets pointed in random directions due to the current and spin of the orbiting electrons
  - In some materials, this intrinsic magnetism is stronger than in others, leading to higher μ.
  - The higher the μ, the more an externally-applied H-field leads to a stronger B-field due to alignment of the atoms and the more magnetic the material is



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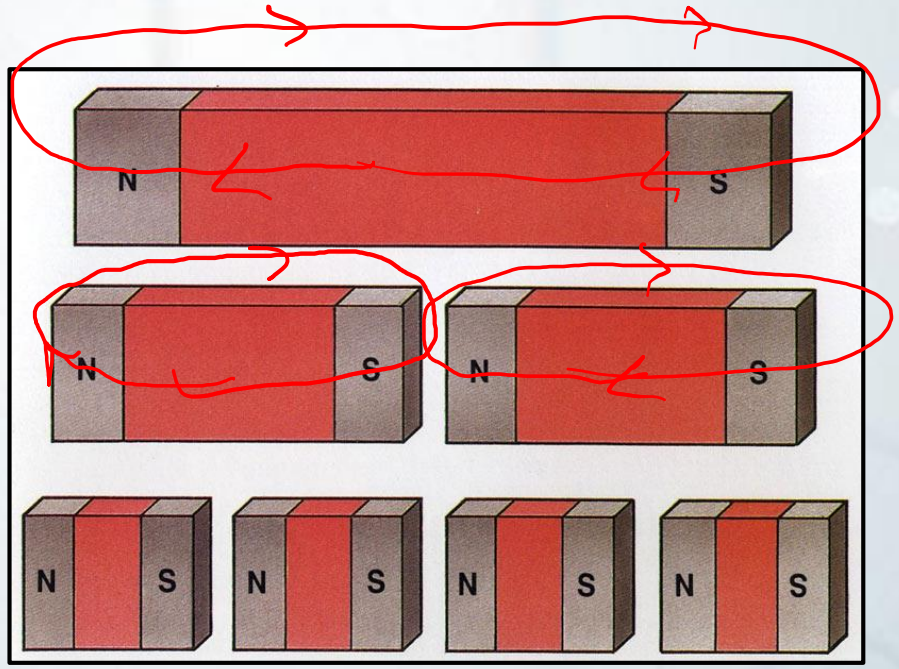
Magnetostatics



Permanent magnets exhibit permanent magnetic fields due to these atomic magnetic moments

Magnetostatics

- A permanent magnet that is cut in two will not separate into "north" and "south" magnetic charges, it will just become two magnets with two new north and south poles.



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- This is due to Maxwell's Equations → "Gauss's Law for magnetic fields"

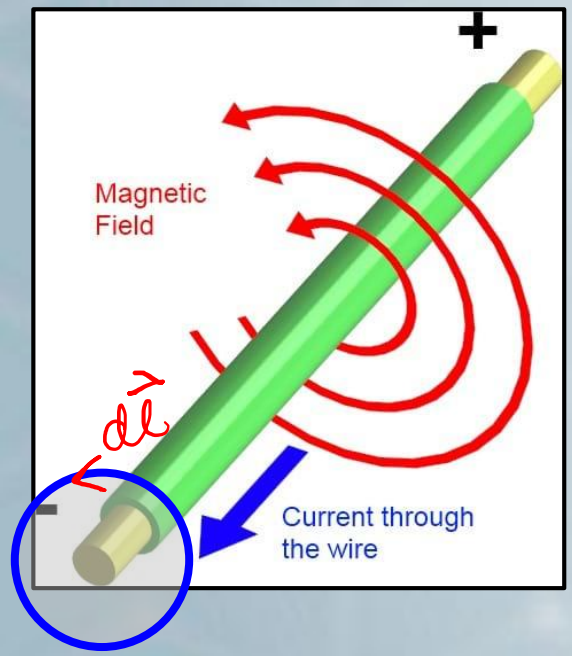
$$\oint \vec{B} \cdot d\vec{S} = 0$$

$$\nabla \cdot \vec{B} = 0$$

Ampere’s Law



Ampere's Law



- In magnetostatics, current is the cause of magnetic field circulation or "whirlpooling".

$$\nabla \times \vec{H} = \vec{J}$$

- We will solve magnetostatic problems by using Ampere's Law and drawing an **Amperian loop**.

- The more current that passes through the loop, the stronger the electric field around the loop

$$\oint \vec{H} \cdot d\vec{l} = \iint \vec{J} \cdot d\vec{s} = I_{enc}$$

Ampere's Law

- In differential form, Ampere's law states:

$$\nabla \times \vec{H} = \vec{J}$$

"Current densities generate a circulating magnetic field according to the right-hand rule"

- Previously, we said the following:

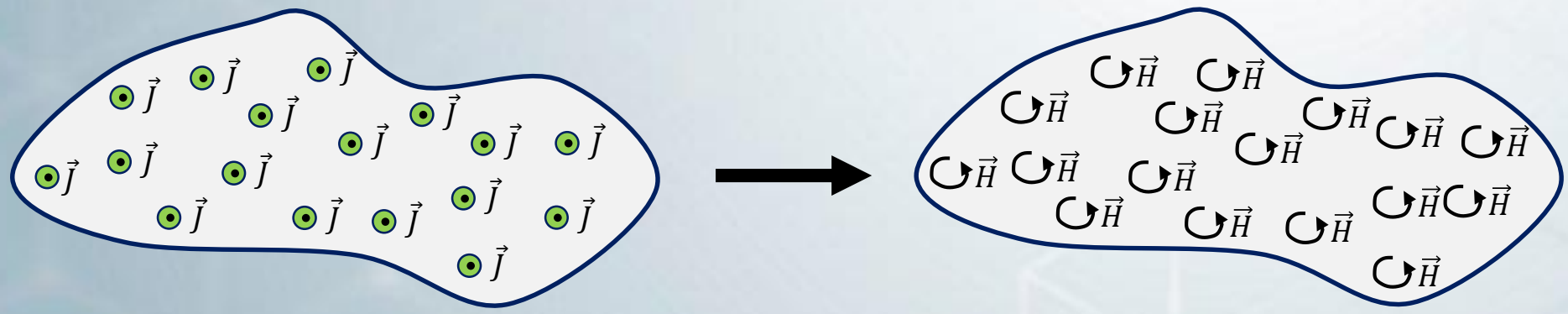


"The circulation of a vector field in practice is often due to a *distribution* of curl - that is, the vector field acts less like it has one whirlpool and more like it has a distribution of infinitesimally-small whirlpools."

How does this relate to Ampere's law?

Ampere's Law

- If we look at the magnetic field generated by multiple local current densities flowing through a surface (coming out of the page):



Each local current density contributes a local magnetic field

Ampere's Law

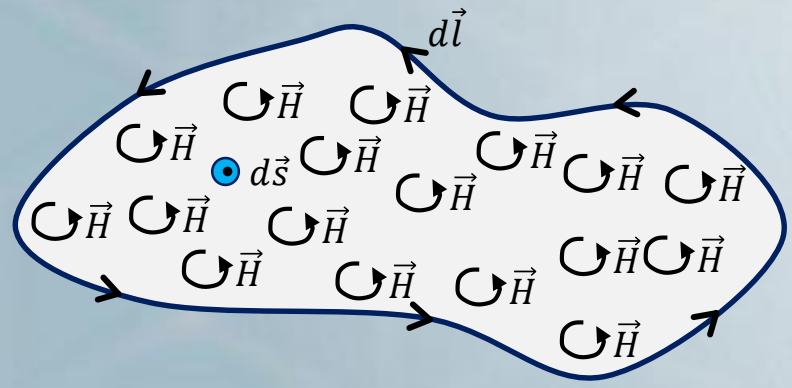
- How do we calculate the resulting magnetic field $\vec{H}$?
- We can use superposition to calculate the net effect of the current density by integrating over the surface:

$$\iint \left(\vec{\nabla} \times \vec{H}\right) \cdot d\vec{s} = \iint \vec{J} \cdot d\vec{s} = I_{enc}$$

- But this doesn't help us much, since solving for $\vec{H}$ inside a double integral with the curl operation applied is difficult. Now what?

Ampere's Law

- Use Stokes' theorem:



$$\iint (\vec{\nabla} \times \vec{F}) \cdot d\vec{s} = \oint \vec{F} \cdot d\vec{l}$$

Assigning $\vec{F} \rightarrow \vec{H}$

$$\iint (\vec{\nabla} \times \vec{H}) \cdot d\vec{s} = \oint \vec{H} \cdot d\vec{l} = I_{enc}$$

Integral form of Ampere's law

Ampere's Law

$d\vec{l}$ and $d\vec{s}$ must correspond to the right-hand rule

thumb: $d\vec{s}$

fingers curl in direction of $d\vec{l}$

Examples of Amperian Loops

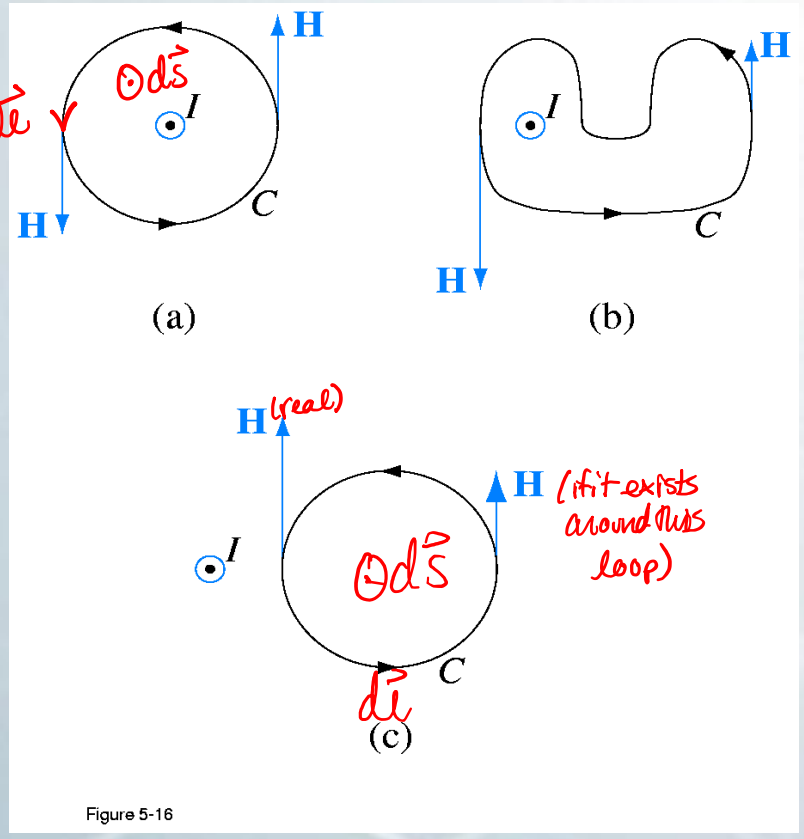
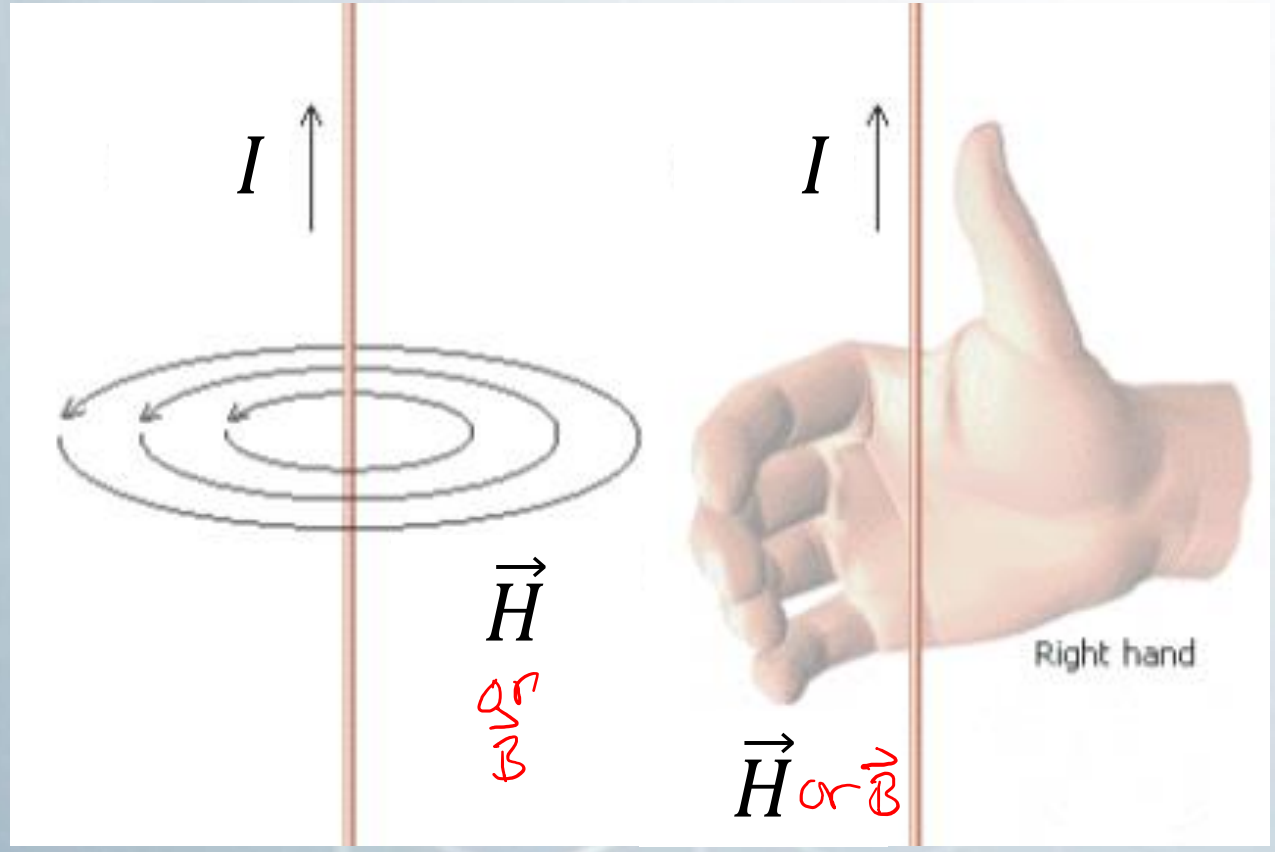


Figure 5-16

$$\oint \vec{H} \cdot d\vec{l} = \iint \vec{J} \cdot d\vec{s} = I_{enc}$$

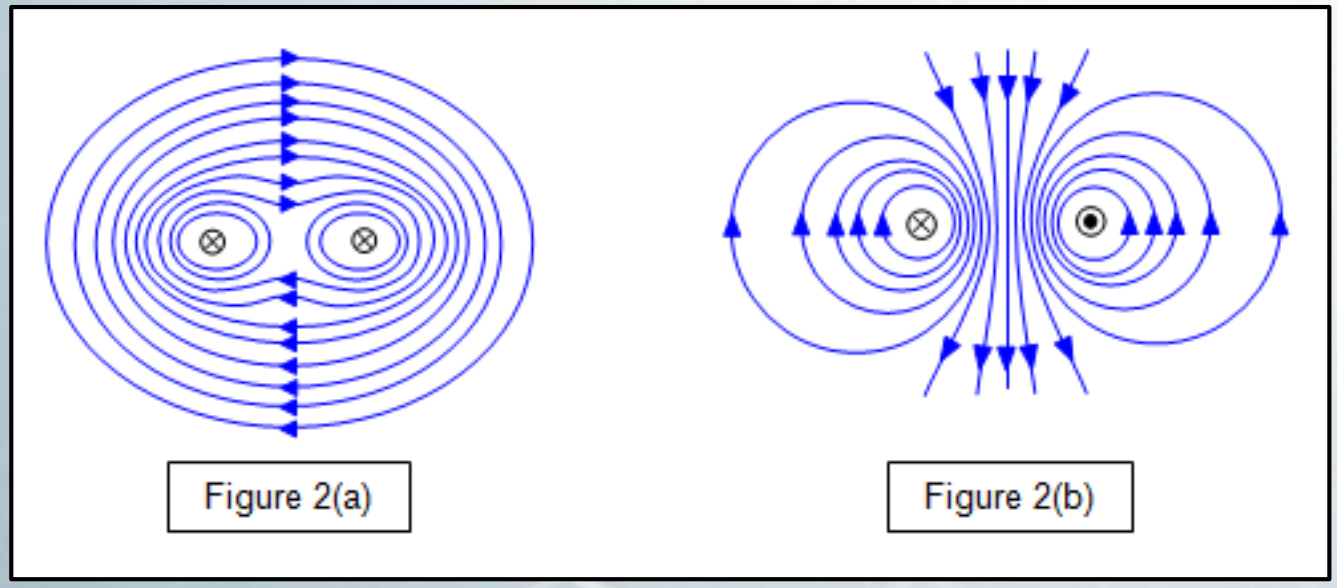
Ampere's Law

- Direction of $\vec{H}$ is given by the right-hand rule



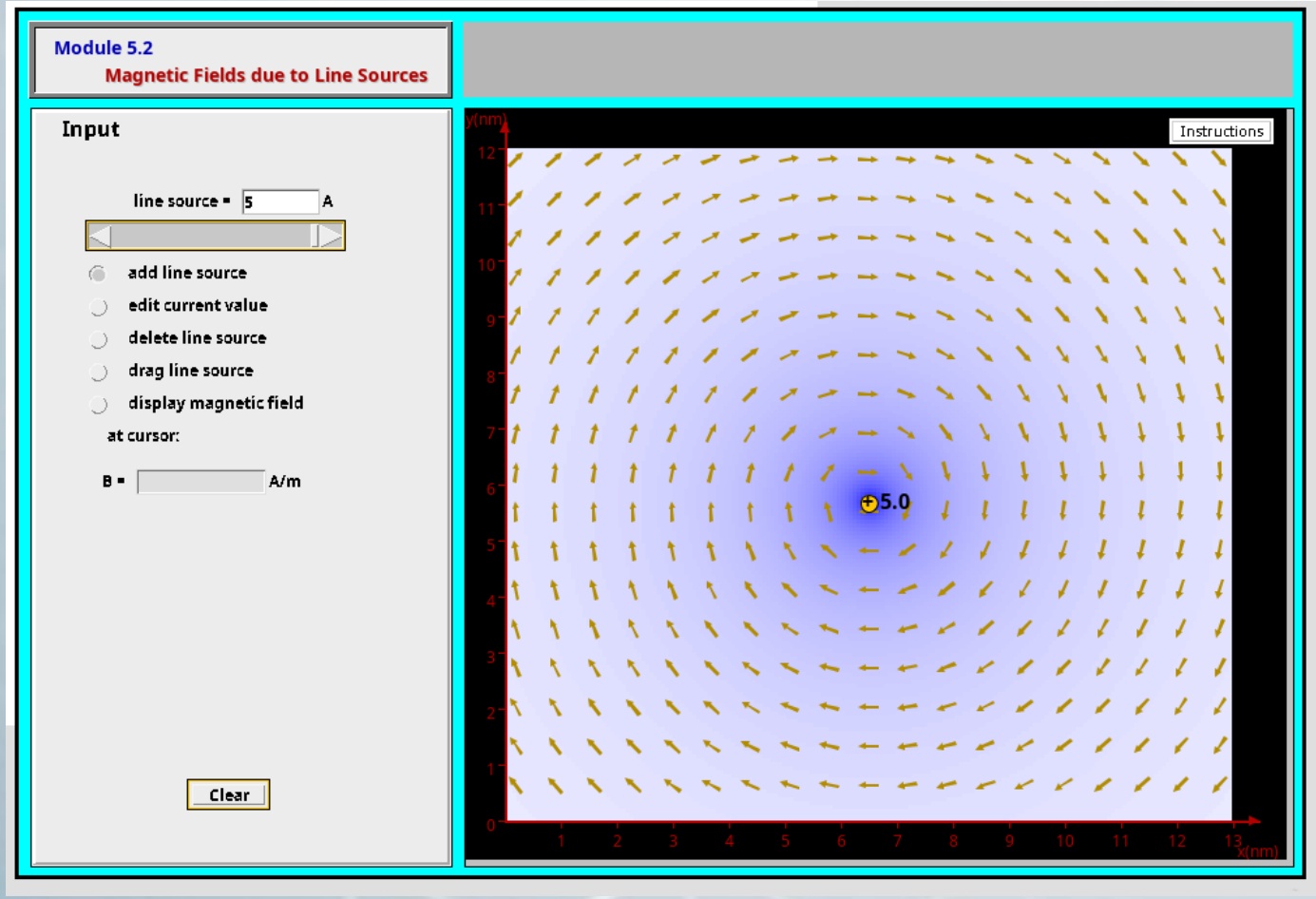
Ampere's Law

- In the case of multiple current channels, use field superposition as you would with point charges in electrostatics.



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Ampere's Law



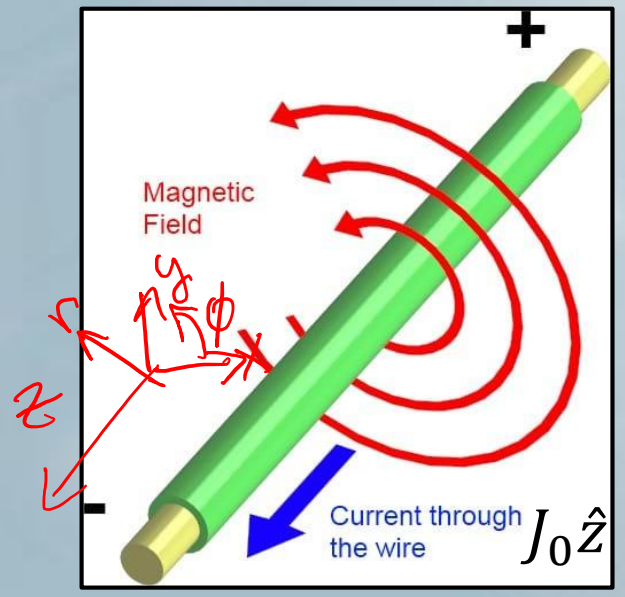
https://em8e.eecs.umich.edu/jsmodules/ch5/mod5_2.html

Ampere's Law

- To begin, let's calculate the magnetic flux density field $\vec{B}$ generated by a wire that carries a current I in the $\hat{z}$ direction

$$\oint \vec{H} \cdot d\vec{l} = \iint \vec{J} \cdot d\vec{s} = I_{enc}$$

$$\vec{B} = \mu \vec{H}$$



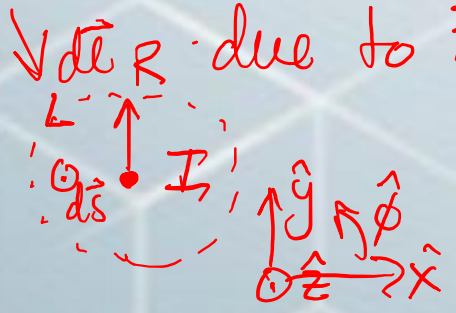
$$\iint (J_0\hat{z}) \cdot (r dr d\phi \hat{r}) = I$$

1. Determine the direction of the H (or B) field using the RHR and, when applicable, superposition and symmetry)

$$\vec{H} = H_\phi(r)\hat{\phi}$$

2. Define an Amperian loop
   a) its surface intersects $\vec{J} \rightarrow \hat{n} \parallel \hat{J}$
   b) the line integral is easy to do $\rightarrow d\vec{\ell} \parallel \hat{H}$

Amperian loop



- due to $\vec{H} \parallel \hat{\phi} \rightarrow d\vec{\ell} \parallel \hat{\phi}$
- due to $\vec{J} \parallel \hat{z} \rightarrow d\vec{s} \parallel \hat{z}$

$$d\vec{s}_z = r\,dr\,d\phi\,\hat{z}$$

$$d\vec{\ell}_\phi = r\,d\phi\,\hat{\phi}$$

Ampere's Law

3. Perform the current integral : $\iint \vec{J} \cdot d\vec{s} = I_{enc}$



$I_{enc} = I$

4. Reform circulation integral : $\oint \vec{H} \cdot d\vec{s} \rightarrow \oint \vec{B} \cdot d\vec{s} = \mu_0 I_{enc}$

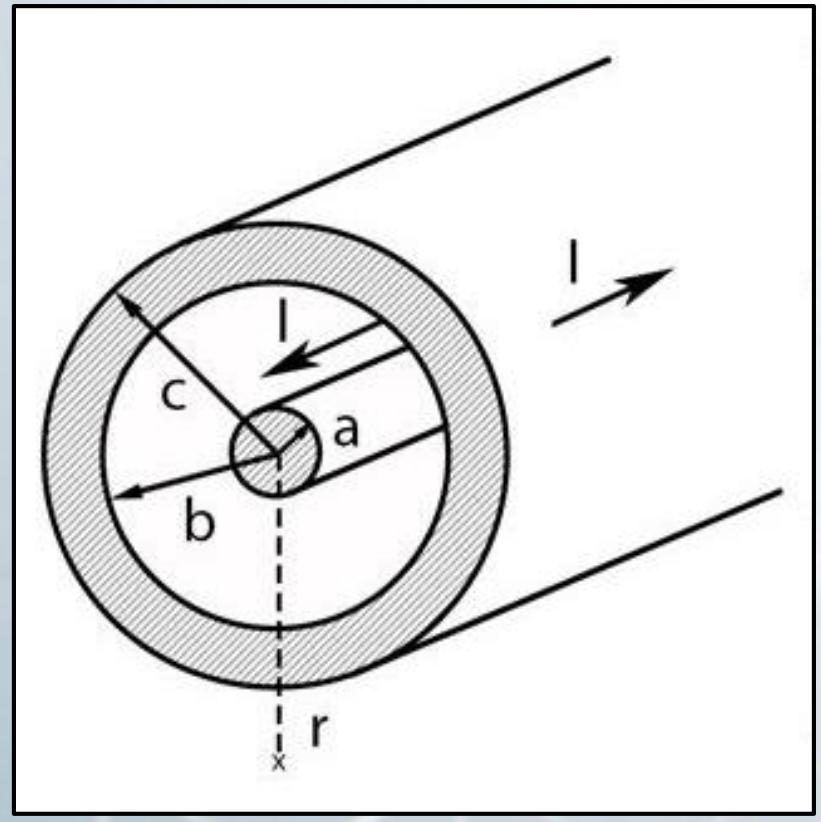
$$\int_0^{2\pi} (B_\phi(r)\hat{\phi}) \cdot (R\,d\phi\,\hat{\phi}) = 2\pi R\, B_\phi(R)$$

5. Equate $\oint \vec{B} \cdot d\vec{s}$ with μI_{enc} to find $\vec{B}$

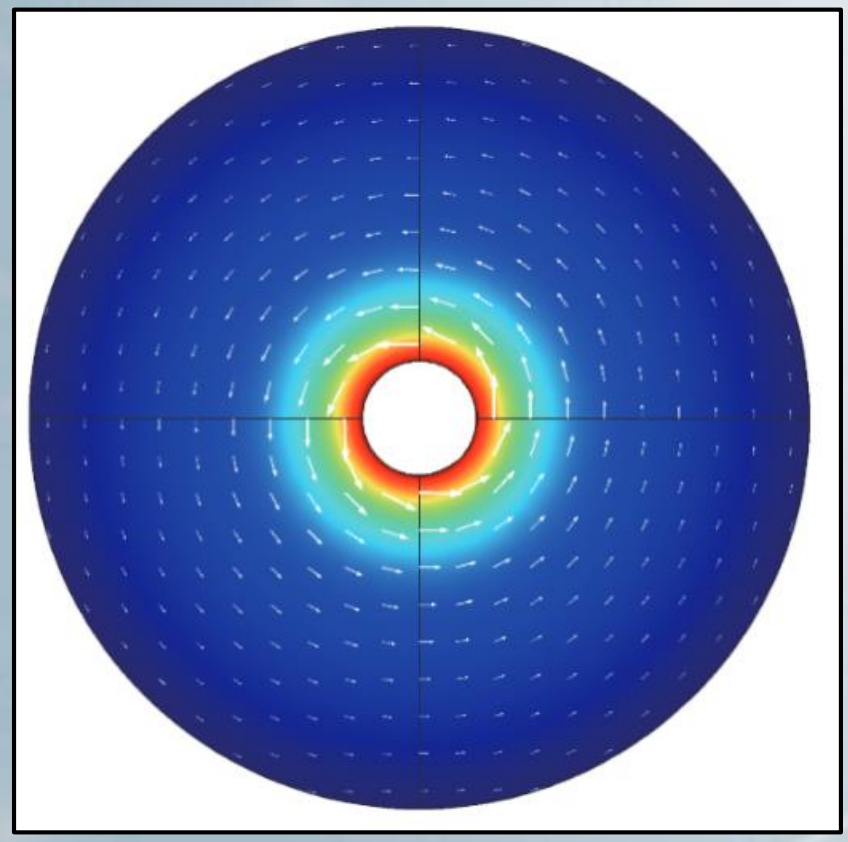
$$2\pi R\, B_\phi(R) = \mu_0 I \Rightarrow \vec{B}(r) = \frac{\mu_0 I}{2\pi r}\hat{\phi}$$

Ampere's Law

- What does the magnetic field look like for this conductive coaxial cable?



Ampere's Law



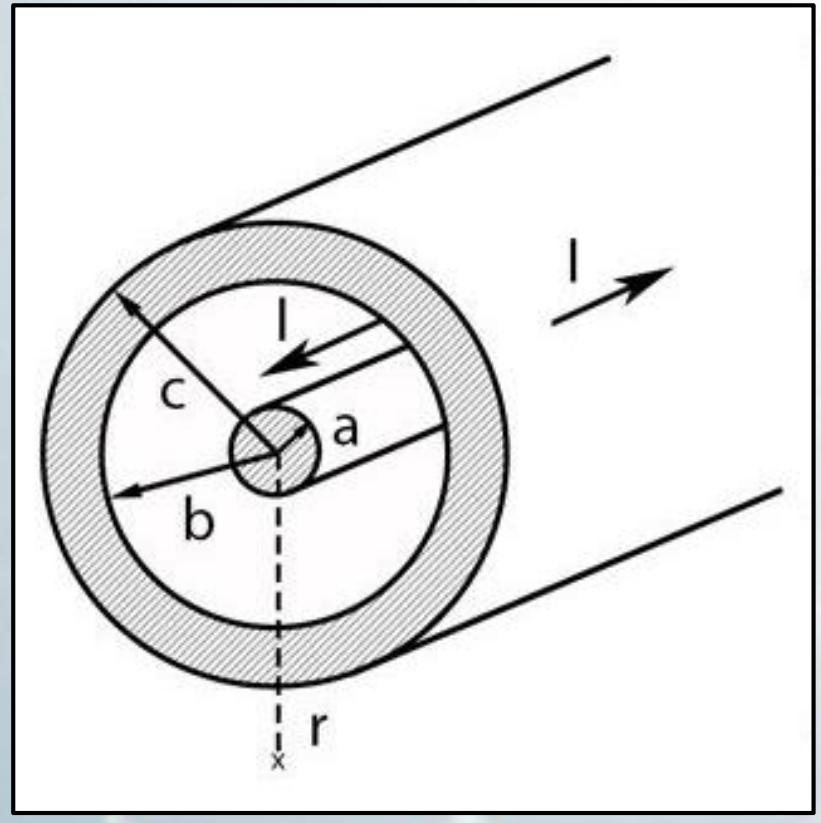
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The field between the conductors looks something like this. (you can check this using the right-hand rule)

A quick point: conductors do not shield magnetic fields in the same way they shield electric fields. Why is this?

Ampere's Law

- Find the magnetic flux density (B) in all regions.



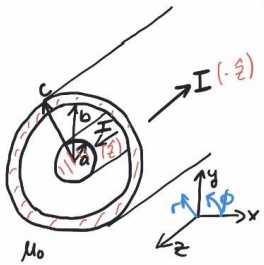
- Assume **J** is constant in the conductors.

Ampere's Law

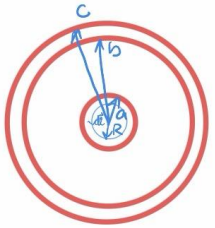
Magnetostatics of a Coaxial Cable

· Find the magnetic flux density field ($\vec{B}$) in all regions. Assume $\vec{J}$ is constant in the conductors and $\mu = \mu_0$ everywhere.

$$\oint \vec{H}\cdot d\vec{\ell} = \iint \vec{J}\cdot d\vec{s} = I_{enc} \rightarrow \oint \vec{B}\cdot d\vec{\ell} = \mu \iint \vec{J}\cdot d\vec{s} = \mu I_{enc}$$



1) Find $\vec{B}$ in the region $0<r\leq a$



By symmetry: $\vec{B} = B_\phi(r)\hat{\phi}$

$$\oint \vec{B}\cdot d\vec{\ell} = \int_0^{2\pi} (B_\phi(r)\hat{\phi})\cdot(R\,d\phi\,\hat{\phi}) = 2\pi R B_\phi(R)$$

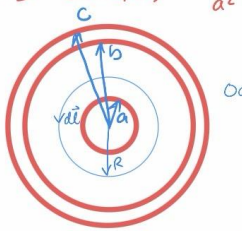
$I_{enc} = \iint \vec{J}\cdot d\vec{s}$ → what is $\vec{J}$?

since $\vec{J}$ is constant: $\vec{J} = \frac{I}{\pi a^2}\hat{z}$

$$I_{enc} = \int_0^{2\pi}\int_0^{R} \left(\frac{I}{\pi a^2}\hat{z}\right)(r\,dr\,d\phi\,\hat{z}) = \frac{I}{\pi a^2}\cdot 2\pi\cdot\frac{R^2}{2} = \frac{I}{a^2}R^2$$

Solve for B: $2\pi R B_\phi(R) = I\frac{R^2}{a^2} \rightarrow \boxed{\vec{B}(r) = \frac{\mu_0 I}{2\pi a^2} r\hat{\phi}}$

2) $\vec{B}$ for the region $a<r\leq b$

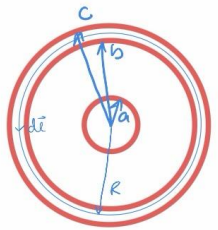


$$\oint \vec{B}\cdot d\vec{\ell} = 2\pi R B_\phi(R)$$

$I_{enc} = I$

$$\boxed{\vec{B}(r) = \frac{\mu_0 I}{2\pi r}\hat{\phi}}$$

3) $\vec{B}$ for the region $b<r\leq c$



$$\oint \vec{B}\cdot d\vec{\ell} = 2\pi R B_\phi(R)$$

↓ from the inner conductor

$I_{enc} = I + \iint \vec{J}_{outer}\cdot d\vec{s}$

$$\vec{J}_{outer} = \frac{I}{\pi c^2 - \pi b^2}(-\hat{z}) = -\frac{I}{\pi(c^2-b^2)}\hat{z}$$

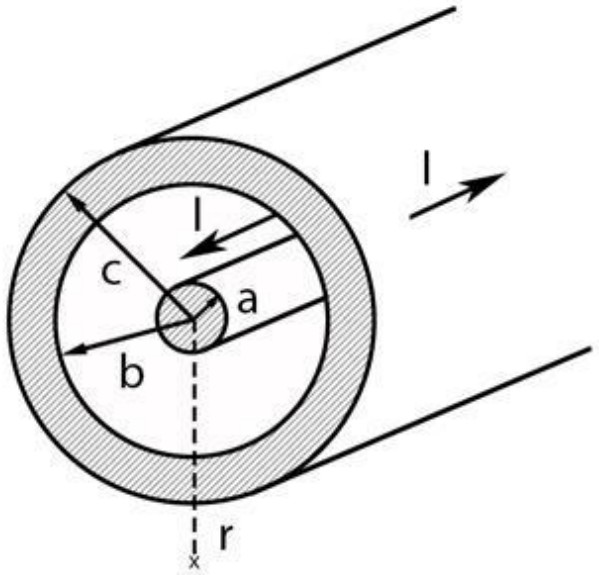
$$I_{enc} = I + \left[\int_0^{2\pi}\int_b^{R}\left(-\frac{I}{\pi(c^2-b^2)}\hat{z}\right)\cdot(r\,dr\,d\phi\,\hat{z})\right]$$

$$= I - \frac{I}{\pi(c^2-b^2)}\cdot 2\pi\,\frac{1}{2}(R^2-b^2) = I\left(1-\frac{R^2-b^2}{c^2-b^2}\right)$$

Solve for B: $\boxed{\vec{B}(r) = \frac{\mu_0 I}{2\pi r}\left(1-\frac{r^2-b^2}{c^2-b^2}\right)\hat{\phi}}$

Ampere's Law

Magnetic Field of a Coaxial Cable



$$B = \frac{\mu_o I}{2\pi a^2} r \qquad (r < a)$$

$$B = \frac{\mu_o I}{2\pi r} \qquad (a < r < b)$$

$$B = \frac{\mu_o I}{2\pi r}\left(\frac{c^2 - r^2}{c^2 - b^2}\right) \qquad (b < r < c)$$

$$B = 0 \qquad (r > c)$$

I : Electric current μ_o : Permeability of free space

B : Magnetic field

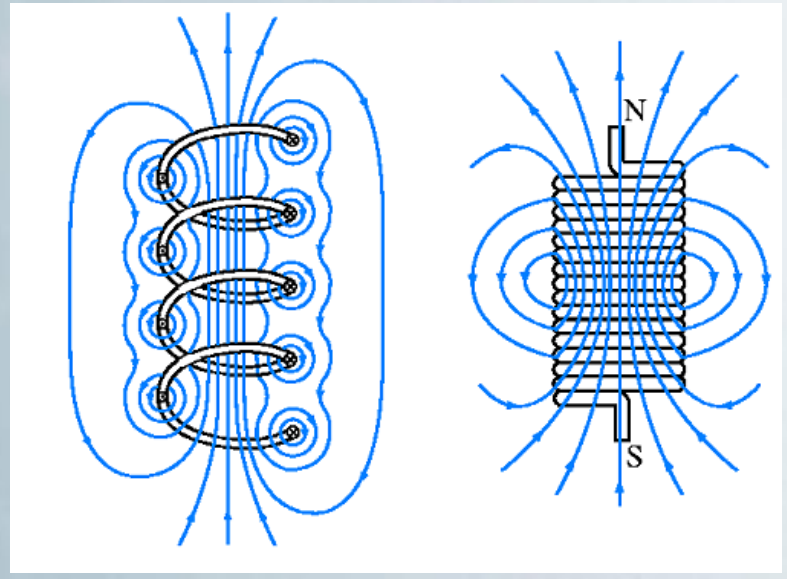
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Ampere’s Law

Do Lecture 15, Exercise 1 in groups of up to 4.

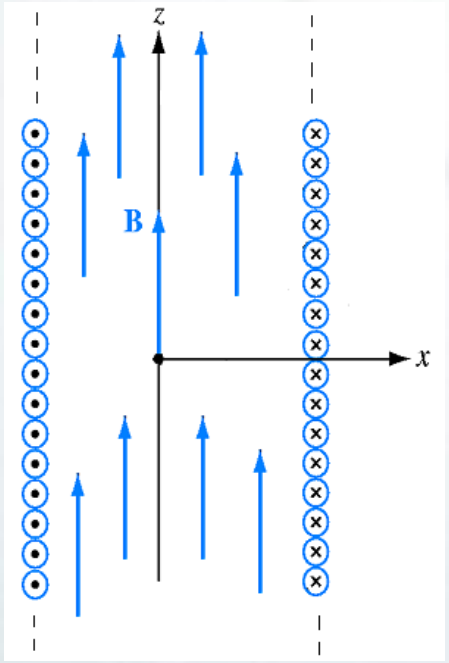
Ampere's Law

Ulaby



(a) Loosely wound solenoid

(b) Tightly wound solenoid

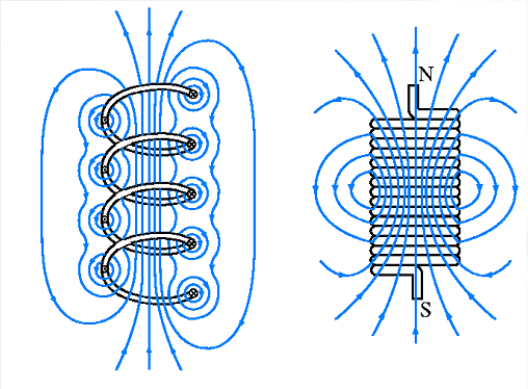


(c) Ideal solenoid

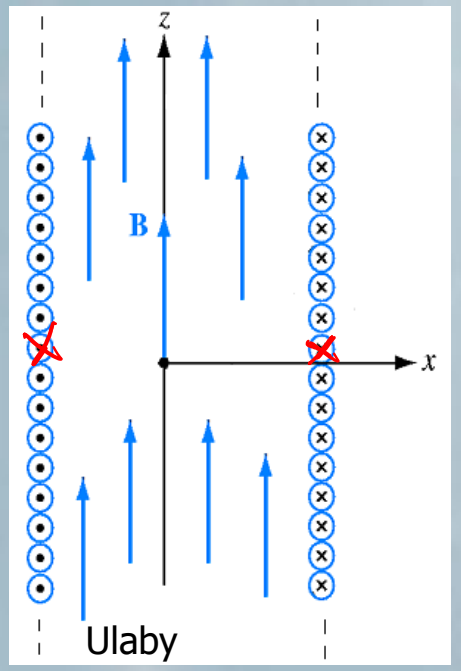
- Just like with Gauss's law, solving problems using Ampere's law requires a great deal of symmetry
- We also use the symmetry of the system to help us choose our Amperian loop geometry, which simplifies the math

Ampere's Law

(a) Loosely wound solenoid



(b) Tightly wound solenoid



(c) Ideal solenoid

- An ideal solenoid has rotational symmetry in ϕ, so the B-field can't change in ϕ:

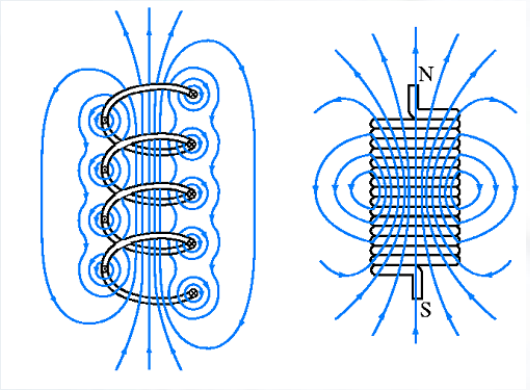
$$\vec{B}(r,\phi,z) = \vec{B}(r,z)$$

- We also know that the B field cannot be parallel to the direction of current flow, so it cannot point in ϕ:

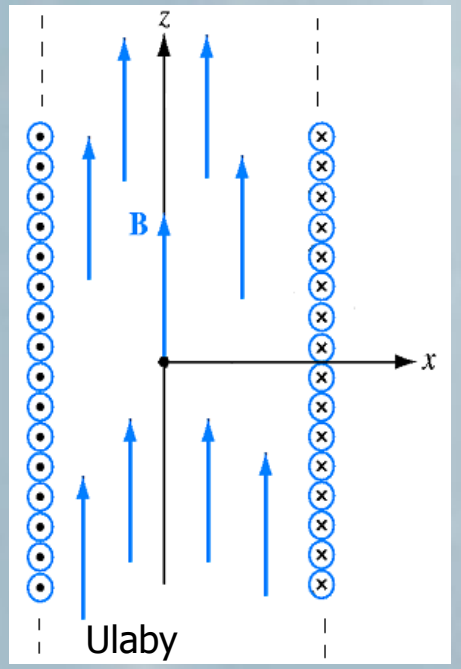
$$\vec{B}(r,z) =< B_r(r,z), 0, B_z(r,z) >$$

Ampere's Law

(a) Loosely wound solenoid



(b) Tightly wound solenoid



(c) Ideal solenoid

- Using the right-hand rule, we know that the field in the plane of a single wire loop also does not point in r:

$$\vec{B}(r,z) = B_z(r,z)\hat{z}$$

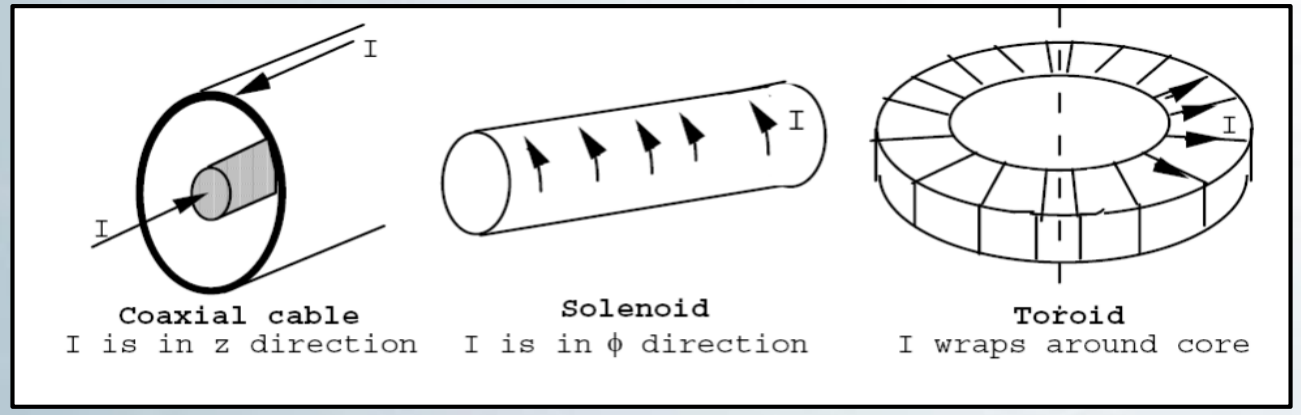
- And because the ideal solenoid is very long, the B-field can't change in z:

$$\vec{B}(r,z) = B_z(r)\hat{z}$$

- Therefore, the B-field points in the $\hat{z}$ direction and varies with changing r:

$$\vec{B}(r) = B_z(r)\hat{z}$$

Ampere's Law



coax $\vec{B} = B_\phi(r)\,\hat{a}_\phi$

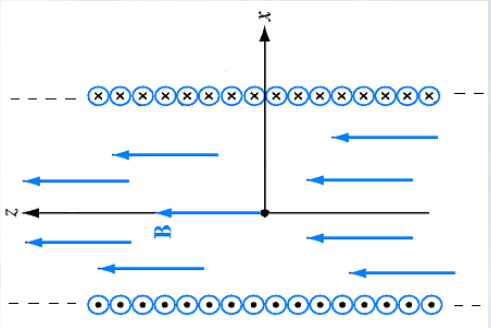
solenoid $\vec{B} = B_z(r)\,\hat{a}_z$

} ignoring end effects

torus $\vec{B} = B_\phi(r,z)\,\hat{a}_\phi$ ← assume tightly wound

Ampere's Law

- What's the basis of the claim that $\vec{B} = 0$ outside the solenoid?



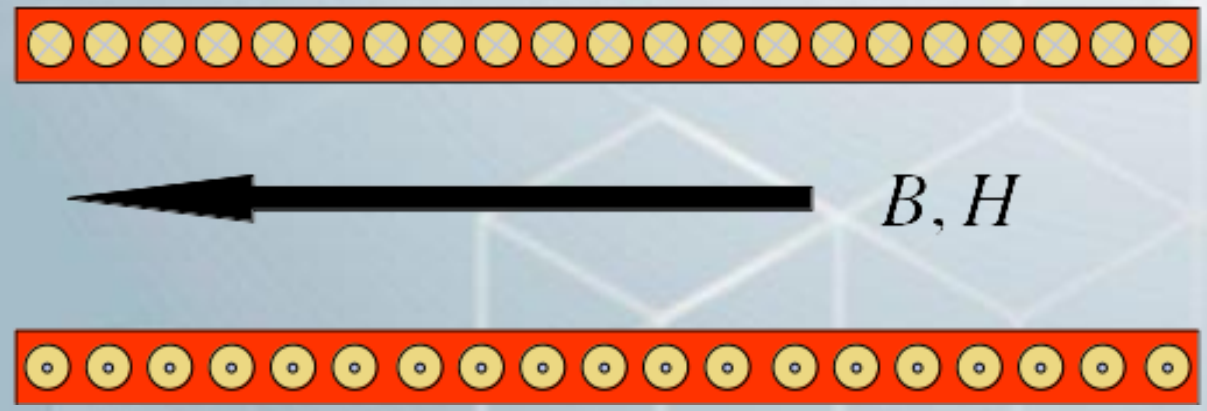
$$\oint H \bullet dl = \int j \bullet ds = I_{net}$$

or

$$\oint B \bullet dl = \mu_0 \cdot \int j \bullet ds = \mu_0 \cdot I_{net}$$

Approach similar to using Gauss' Law, use symmetry to get B-field out of integral

Example: Consider an infinite wire solenoid

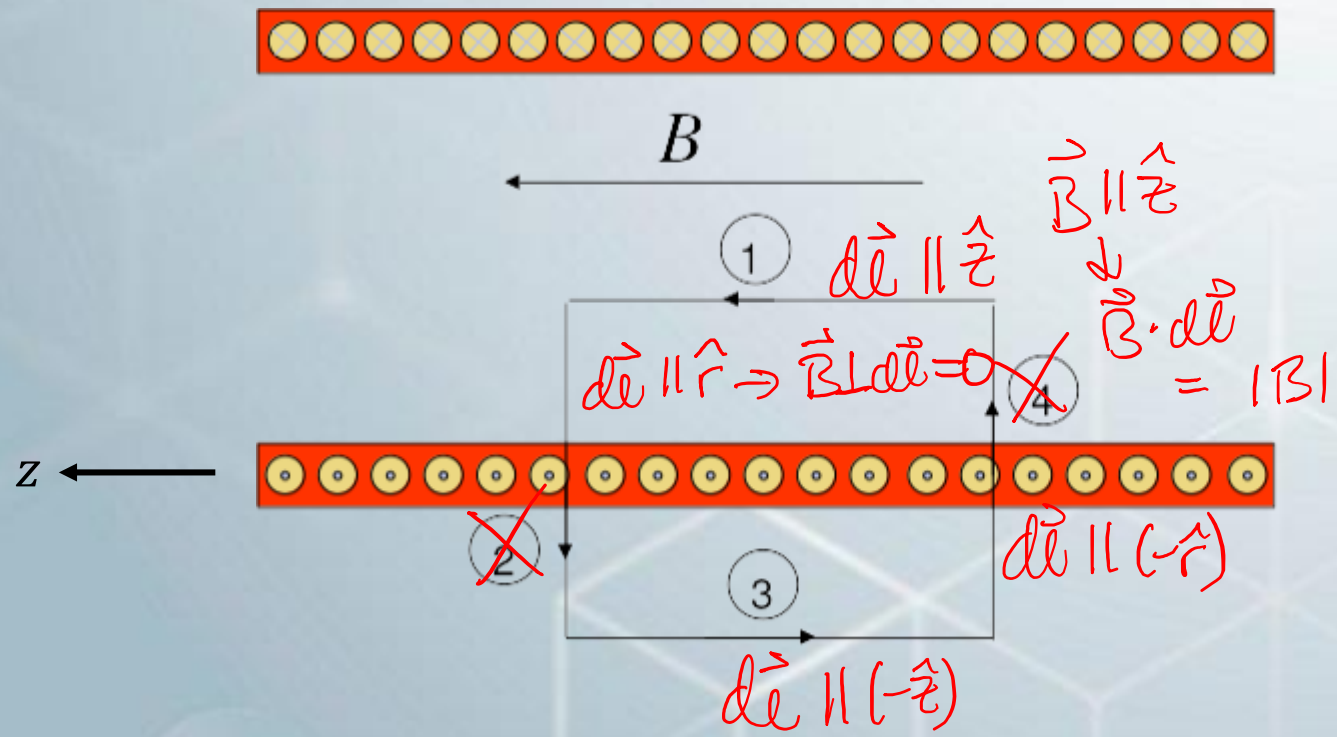


Ampere's Law

- What's the basis of the claim that $\vec{B} = 0$ outside the solenoid?

STEP 1: Choose path for integral - $\oint B \bullet dl$

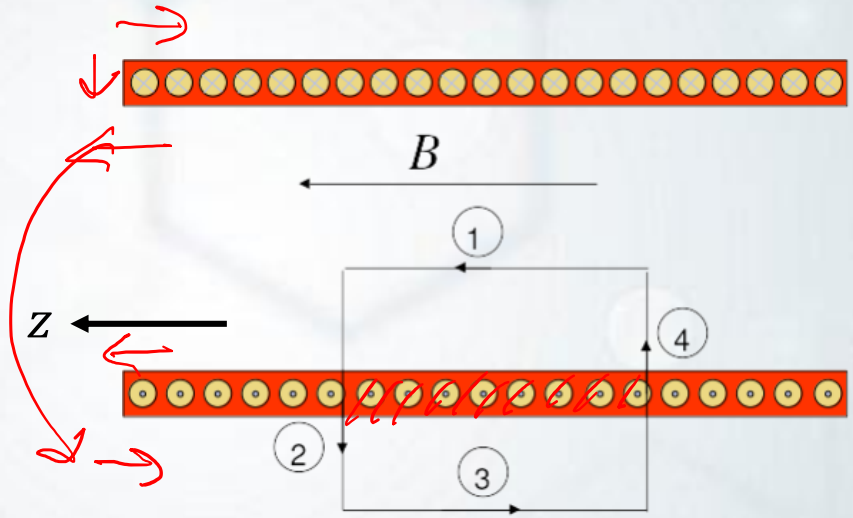
- Chosen paths are 1,2, 3 and 4 - they form a closed loop



Find B

Solenoid has current I through **n** turns/length

Ampere's Law



STEP 2: Evaluate $\oint B \bullet dl$

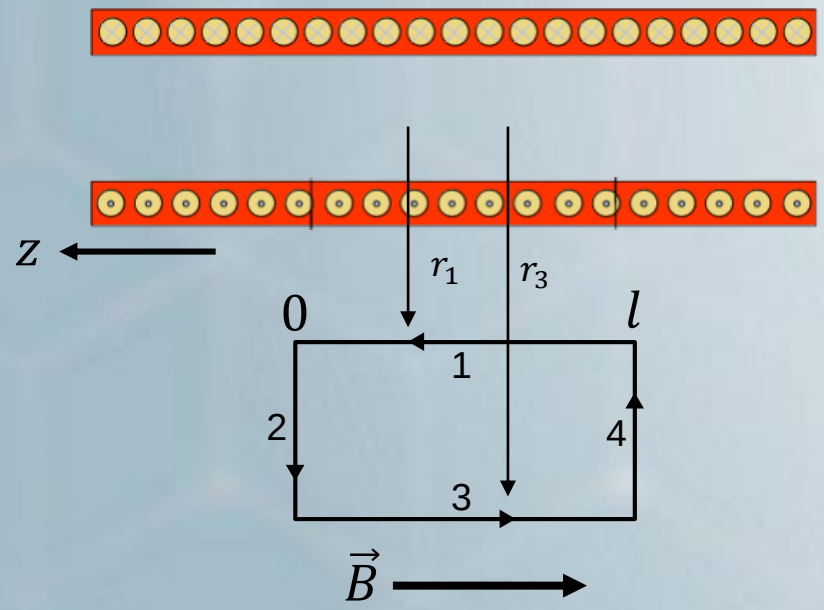
- Segments 2 and 4 have $B \perp dl$

$$\therefore \oint B \bullet dl \Rightarrow 0$$

- Segment 3 has $B \rightarrow 0$ (will show later)
- Segment 1 has $\oint \vec{B} \bullet \vec{dl} = \int_0^l B_z \cdot dz = B_z \cdot l$ → arbitrary length

Ampere's Law

- What's the basis of the claim that $\vec{B} = 0$ outside the solenoid?



Ampere's Law

STEP 3: find I_{net}

- current passing through loop :

$$I_{net} = n \cdot l \cdot I$$

length & current / turn

turns / length

STEP 4: solve for B

$$\oint B \bullet dl = \mu_0 \cdot \int j \bullet ds = \mu_0 \cdot I_{net}$$

$$\Rightarrow \quad B_z \cdot l = \mu_0 \cdot n \cdot l \cdot I \quad \Rightarrow \quad B = \mu_0 \cdot n \cdot I \cdot \hat{a}_z$$

Ampere’s Law

Do Lecture 15, Exercise 2 in groups of up to 4.

Wrap-Up

- Exam 2 next Monday. Covers Lectures 8 - 14
- No calculators, just fun!